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|-----------------------------------------------------------------------------------|------------------------------------------------------------------|--------------|----------------------------------------------------------|------------------|-------------------|
|  |                                                                  | Course Name  | <b>Power Plant Electrical Equipment (MPA-ELE 25/26Z)</b> |                  |                   |
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|                                                                                   |                                                                  | Year         | <b>2025</b>                                              | Study group      | <b>1</b>          |
|                                                                                   |                                                                  | Collaborator | <b>Isgandar, Alan</b>                                    | Measurement Date | <b>03/11/2025</b> |
| Revised by                                                                        |                                                                  | Final Mark   | Exam Date                                                |                  |                   |
| Task Number                                                                       | Task Name                                                        |              |                                                          |                  |                   |
| <b>A</b>                                                                          | <b>Measuring synchronous generator no-load and short-circuit</b> |              |                                                          |                  |                   |

## Assignment

The objectives of this laboratory exercise were:

- To perform a no-load (open-circuit) test on a synchronous generator by measuring terminal voltage as a function of excitation current.
- To perform a short-circuit test on a synchronous generator by measuring stator current as a function of excitation current.
- To use the data from these tests to understand the generator's characteristics and determine its longitudinal synchronous reactance ( $X_d$ )
- To calculate the nominal reactance ( $X_n$ ) from the machine's nameplate data and determine the relative synchronous reactance ( $x_d$ )
- To perform the practical procedure for safely synchronizing the generator and connecting it to the live electrical grid.

## Theoretical introduction

A synchronous generator is an electrical machine that converts mechanical energy into alternating-current (AC) electrical energy. Its operation relies on a rotating magnetic field (produced by the rotor, or "exciter") inducing a voltage in stationary conductors (the stator windings).

**1. No-Load (Open-Circuit) Characteristic (OCC)** This test determines the relationship between the rotor's excitation current ( $I_B$ ) and the stator's open-circuit terminal voltage ( $U_0$ ). The generator is driven at its nominal synchronous speed with the stator terminals disconnected from any load. As the excitation current  $I_B$  is increased, the magnetic flux increases, which in turn increases the induced EMF (voltage).

Initially, this relationship is linear (the "air-gap line") as the magnetic circuit is unsaturated. At higher excitation currents, the iron core begins to saturate, and the voltage increase becomes non-linear; a larger increase in  $I_B$  is required for a small increase in  $U_0$ .

**2. Short-Circuit Characteristic (SCC)** This test determines the relationship between the excitation current ( $I_B$ ) and the stator current ( $I_K$ ) when the stator terminals are symmetrically short-circuited. The generator is again driven at synchronous speed.

Even with a short-circuit, the stator current is limited by the machine's internal impedance, which is primarily its synchronous reactance ( $X_d$ ). Because the stator current creates a strong demagnetizing armature reaction, the net magnetic flux in the machine is very low, and the iron core does not saturate. Therefore, the short-circuit characteristic ( $I_K$  vs  $I_B$ ) is a straight line.

**3. Synchronous Reactance ( $X_d$ ).** The longitudinal synchronous reactance ( $X_d$ ) is a critical parameter for modelling the generator. It is determined from the OCC and SCC data. For a specific excitation current ( $I_B$ ), the unsaturated synchronous reactance is the ratio of the no-load voltage (from the linear air-gap line) to the short-circuit current (from the SCC,  $I_K$ ) at that same excitation current.

$$X_d = U_0 / I_K$$

**4. Grid Synchronization** Connecting a generator to the main electrical grid (an "infinite bus") requires a precise synchronization process to prevent severe electrical and mechanical damage. The following conditions must be met:

1. **Equal Voltages:** The generator's RMS terminal voltage must be equal to the grid's RMS voltage.
2. **Equal Frequencies:** The generator's frequency must be equal to the grid's frequency.
3. **Same Phase Sequence:** The phase sequence of the generator must match the grid.
4. **Zero Phase Angle:** At the moment of connection, the generator's voltage waveform must be in phase with the grid's voltage waveform.

This process is monitored using a synchroscope or, as in this lab, a set of voltmeters.

# Procedure

The lab work was conducted in three main parts:

## 1. No-Load Test:

The generator's prime mover was activated and brought to the nominal synchronous speed. With the stator terminals open-circuited, the DC excitation current ( $I_b$ ) was slowly increased in steps from 0 A up to 1.2 A. At each step, the excitation current and the three phase-to-phase terminal voltages ( $U_{of1}$ ,  $U_{of2}$ ,  $U_{of3}$ ), were recorded. The excitation current was then slowly decreased in steps from 1.2 A back to 0 A, and the voltages were recorded again to observe the effects of magnetic hysteresis. The average voltage was calculated at each step.

Measurement of no-load characteristics

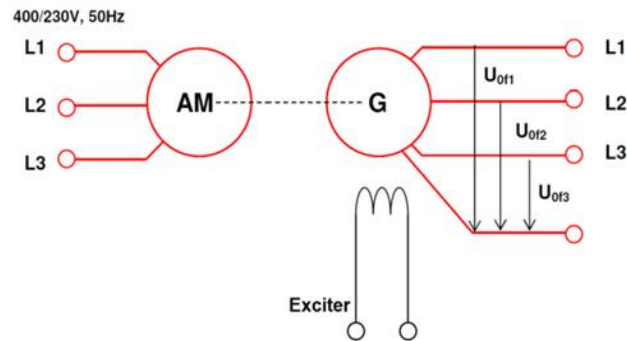
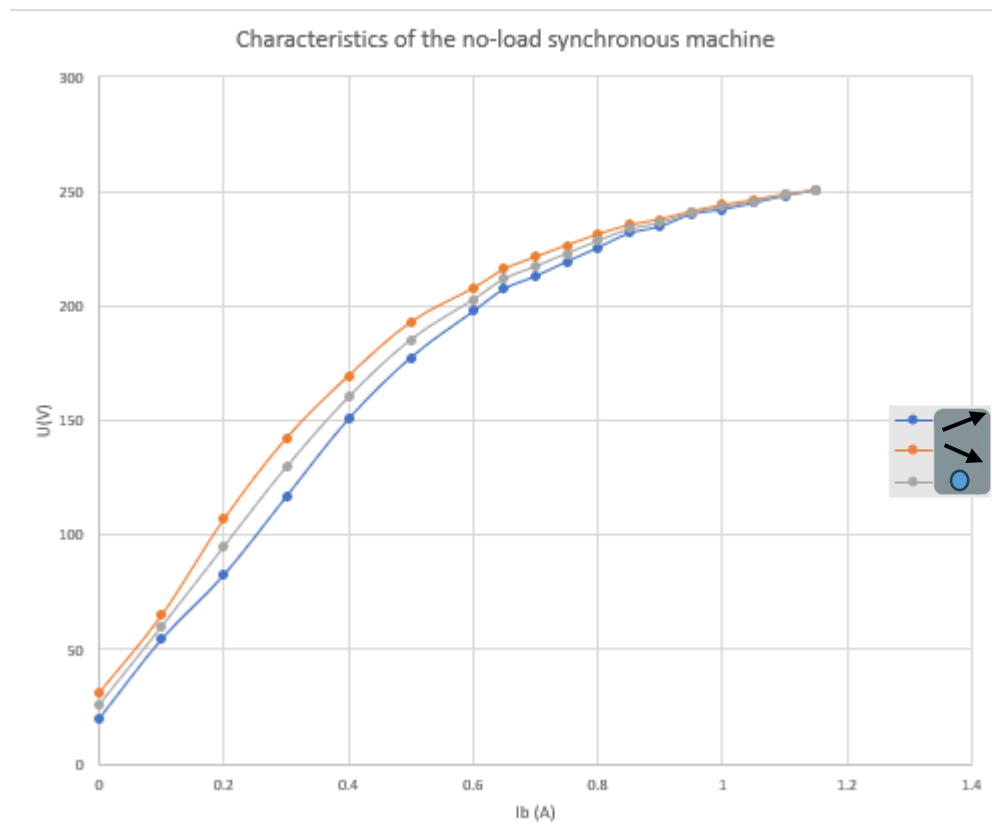


Fig. 1 Wiring synchronous machine at no-load

After performing the measurements, we noted the voltage values for each phase at each step.

| Excitation current<br>[A] | No load |        |        |           |        |        |        |           |          |
|---------------------------|---------|--------|--------|-----------|--------|--------|--------|-----------|----------|
|                           | Up      |        |        |           | Down   |        |        |           | Uavg [V] |
|                           | U1 [V]  | U2 [V] | U3 [V] | Uavg. [V] | U1 [V] | U2 [V] | U3 [V] | Uavg. [V] |          |
| 0                         | 19.8    | 19.8   | 19.8   | 19.8      | 31.6   | 31.6   | 31.6   | 31.6      | 25.7     |
| 0.1                       | 54.2    | 54.6   | 54.5   | 54.4333   | 65.6   | 65     | 65     | 65.2      | 59.8167  |
| 0.2                       | 82.4    | 82.6   | 82.6   | 82.5333   | 107.2  | 107    | 107.1  | 107.1     | 94.8167  |
| 0.3                       | 116.9   | 116.6  | 116.7  | 116.733   | 142.5  | 142.3  | 142.3  | 142.367   | 129.55   |
| 0.4                       | 150.9   | 150.7  | 150.6  | 150.733   | 169.9  | 169.5  | 169.5  | 169.633   | 160.183  |
| 0.5                       | 177.8   | 177.6  | 177.4  | 177.6     | 193.1  | 193    | 193.1  | 193.067   | 185.333  |
| 0.6                       | 197.9   | 197.6  | 197.8  | 197.767   | 208    | 207.8  | 207.7  | 207.833   | 202.8    |
| 0.65                      | 207.8   | 207.6  | 207.5  | 207.633   | 216.3  | 216.2  | 216    | 216.167   | 211.9    |
| 0.7                       | 213.3   | 213.2  | 213    | 213.167   | 221.5  | 221.4  | 221.4  | 221.433   | 217.3    |
| 0.75                      | 219.4   | 219.4  | 219.7  | 219.5     | 226.7  | 226.5  | 226.5  | 226.567   | 223.033  |
| 0.8                       | 225.7   | 225.6  | 225.5  | 225.6     | 231.5  | 231.3  | 231.2  | 231.333   | 228.467  |
| 0.85                      | 232     | 231.9  | 231.8  | 231.9     | 235.5  | 235.3  | 235.2  | 235.333   | 233.617  |
| 0.9                       | 235     | 235    | 234.8  | 234.933   | 238    | 237.8  | 237.8  | 237.867   | 236.4    |
| 0.95                      | 240.1   | 240.3  | 240.3  | 240.233   | 241.5  | 241.3  | 241.3  | 241.367   | 240.8    |
| 1                         | 242.3   | 242.2  | 242.4  | 242.3     | 244.3  | 244    | 244    | 244.1     | 243.2    |
| 1.05                      | 245.1   | 245.3  | 245    | 245.133   | 246.3  | 246.4  | 246.2  | 246.3     | 245.717  |
| 1.1                       | 248.2   | 248.5  | 248.4  | 248.367   | 248.9  | 248.7  | 248.6  | 248.733   | 248.55   |
| 1.15                      | 250.9   | 250.8  | 250.6  | 250.767   | 250.6  | 250.4  | 250.5  | 250.5     | 250.633  |

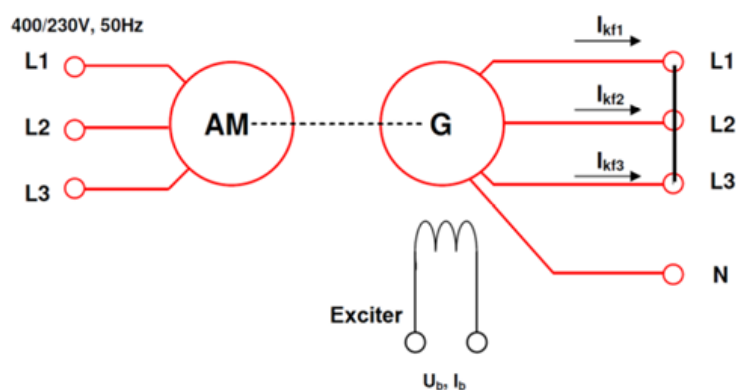
Based on the data we get from the measurements we plotted them to the graphic to observe characteristics of the no-load synchronous machine.



## 2. Short-Circuit Test:

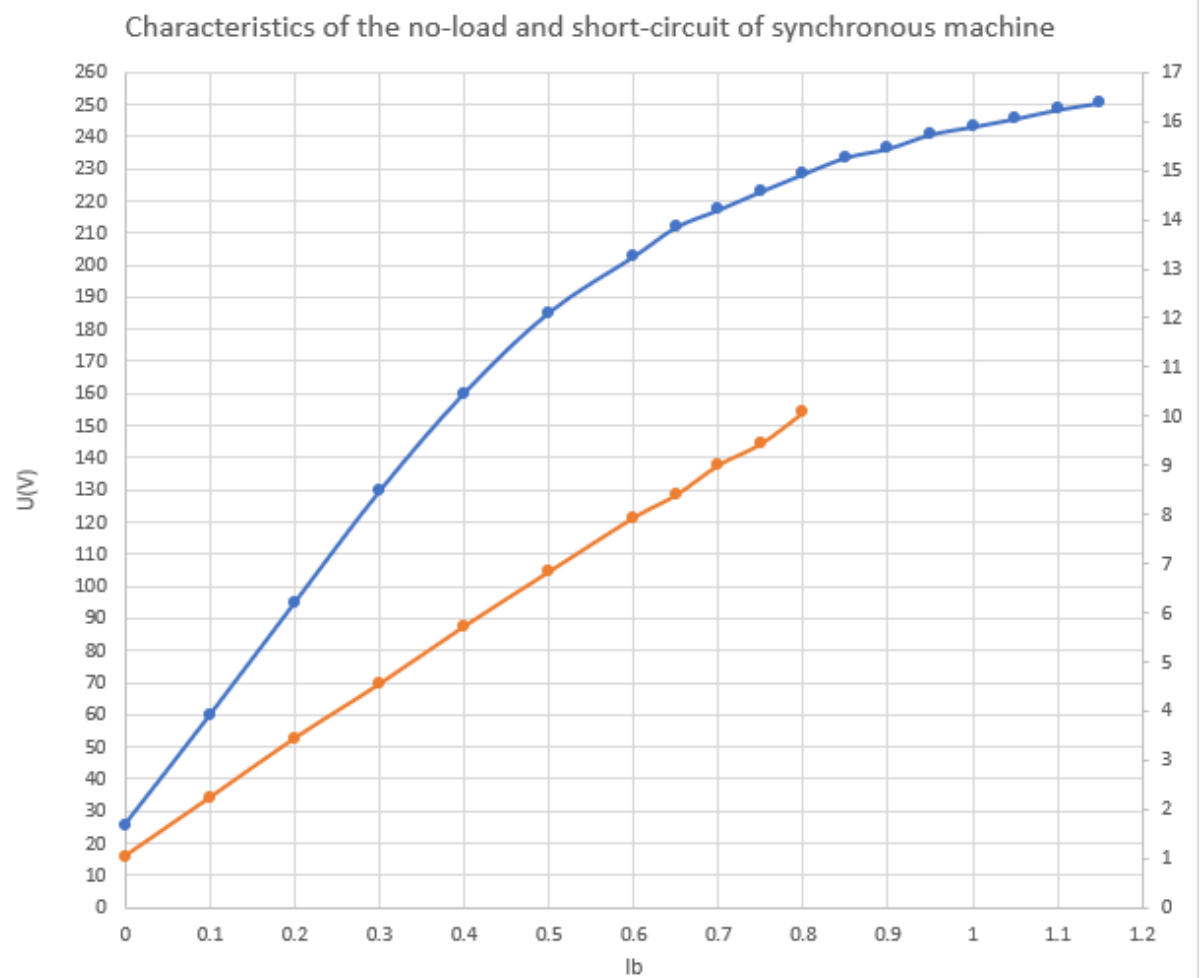
The generator was again brought to synchronous speed. The stator terminals were connected to a three-phase short-circuit block via ammeters. The excitation current ( $I_b$ ) was *very carefully* and slowly increased from 0 A. At each step, the excitation current and the three short-circuit phase currents ( $I_{kf1}$ ,  $I_{kf2}$ ,  $I_{kf3}$ ) were recorded. The excitation was increased until the stator current reached a safe maximum (e.g., nominal current) or the excitation limit (e.g., 750 mA) was reached. The average short-circuit current was calculated at each step.

### Measurement of short circuit characteristics



As we did it in the no-load circuit, for this step we measured the current values in short-circuit and calculated the average for each step:

| Short circuit |        |        |           |
|---------------|--------|--------|-----------|
| I1 [A]        | I2 [A] | I3 [A] | Iavg. [A] |
| 1.04          | 1.05   | 1.03   | 1.04      |
| 2.23          | 2.24   | 2.23   | 2.23333   |
| 3.45          | 3.45   | 3.43   | 3.44333   |
| 4.55          | 4.56   | 4.54   | 4.55      |
| 5.73          | 5.73   | 5.72   | 5.72667   |
| 6.85          | 6.84   | 6.83   | 6.84      |
| 7.95          | 7.95   | 7.93   | 7.94333   |
| 8.42          | 8.41   | 8.39   | 8.40667   |
| 9.02          | 9.02   | 9.01   | 9.01667   |
| 9.46          | 9.45   | 9.44   | 9.45      |
| 10.09         | 10.1   | 10.08  | 10.09     |



### 3. Synchronization and Grid Connection:

The final part of the exercise was to connect the generator to the live grid.

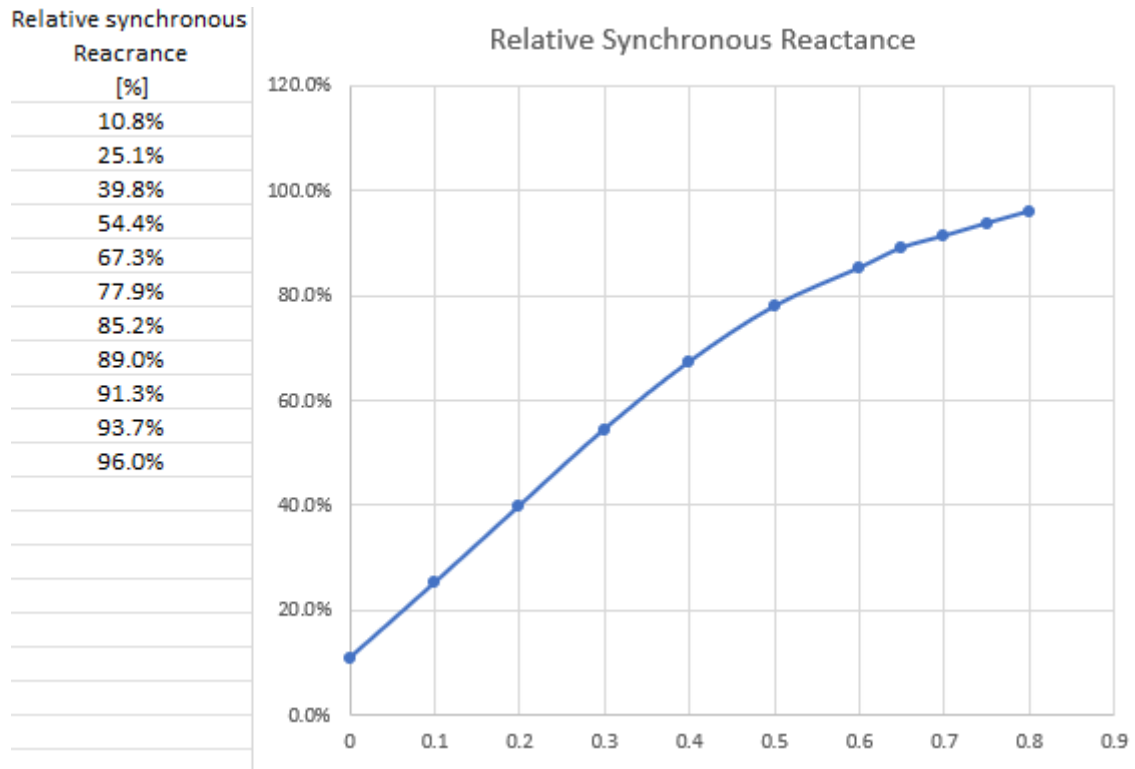
1. The generator was run at synchronous speed, and its excitation was adjusted so that its terminal voltage matched the grid voltage.
2. Three manual voltmeters were used as a synchroscope, connected between the generator terminals and the grid terminals, to monitor the voltage and phase difference.
3. Initially, the frequencies were set to 50 Hz and were the same. The voltmeters showed a stable value.
4. To observe the synchronization process, the frequency of the generator (controlled by its prime mover) was changed. This frequency difference between the generator and the grid created a "beating" effect.
5. We observed that the voltmeter values started changing, or fluctuating. The inspector noted this was the result of the frequency difference creating a constantly changing phase difference between the generator and the grid. The speed of the fluctuation indicated the slip frequency (the difference between the two frequencies).
6. The generator's frequency was carefully adjusted to be very close to the grid frequency, slowing the fluctuation. We then waited for the moment when all three voltmeters came to 0. This "dark" moment indicated that the generator's voltage and the grid's voltage were momentarily in phase (zero phase difference).
7. At that precise instant, the circuit breaker was closed, connecting the generator to the grid.
8. It was observed that even though the generator's frequency was slightly different before connection, as soon as it was connected, it "obeyed" the system frequency. The powerful grid (infinite bus) locked the generator's speed into perfect synchronism.

### 4. Nominal and Relative Synchronous Reactance ( $X_n$ and $x_d$ )

**Relative Synchronous Reactance ( $x_d$ ):** This is the per-unit value of the synchronous reactance, defined as  $x_d = X_D / X_n$ . A related characteristic, which is plotted in this lab to show the machine's saturation, is calculated from the no-load curve. This value is found by taking the measured no-load phase voltage ( $U_0$ ) at a given excitation current ( $I_b$ ), dividing it by the nominal current ( $I_n$ ), and then normalizing this pseudo-impedance by the nominal impedance ( $X_n$ ).

$$X_d(I_b) = (\underline{U_0})(I_b) / (I_n) * (X_n)$$

This function,  $(x_d) = f(I_b)$  is not a constant value but rather a curve that reflects the magnetic saturation. As  $I_b$  increases,  $U_o$  increases, but at a decreasing rate due to saturation, which is reflected in the shape of this curve.



## List of instruments used

The following equipment was used in the experiment. (Note: Specific models, serial numbers, and accuracy classes should be recorded from the lab equipment for a complete report).

- Synchronous Generator (Type, S/N, Nameplate Data)
- Prime Mover (e.g., DC Motor)
- DC Power Supply (for excitation)
- AC Voltmeters (3) (for no-load test and synchronization)
- AC Ammeters (3) (for short-circuit test)
- DC Ammeter (for excitation current)
- Tachometer (to verify synchronous speed)
- Synchronization Module (or three-voltmeter setup)
- Circuit Breaker and Connection Leads

## Conclusion

The no-load and short-circuit tests were successfully performed, and measurements were recorded. This data is sufficient for plotting the Open-Circuit Characteristic (OCC) and Short-Circuit Characteristic (SCC) curves. The OCC plot would be expected to show the effect of magnetic saturation, while the SCC plot would be expected to be linear.

From these two curves, the longitudinal synchronous reactance ( $X_d$ ) can be calculated, which is a fundamental parameter for modelling the generator's behaviour under load.

The synchronization procedure was also completed successfully. The use of the three-voltmeter method provided a clear visual demonstration of the "beating" phenomenon caused by a frequency difference between the generator and the grid. The fluctuating voltmeter readings were correctly identified as the result of the shifting phase difference. The successful connection at the zero-volt (in-phase) moment confirmed the correct application of synchronization theory. The generator's immediate "pull-in" to the grid frequency upon connection vividly demonstrated the concept of the infinite bus.